

SAFETY DATA SHEET

SECTION 1 IDENTIFICATION

1.1 PRODUCT IDENTIFIER	477 / HR-477
1.2 OTHER MEANS OF IDENTIFICATION	Sulfuric Acid Titrating Solution (1drop = 10ppm-50ppm/25ml sample)
1.3 RECOMMENDED USE / RESTRICTIONS ON USE	Analytical Testing Reagent
1.4 SUPPLIERS DETAILS	WATER SCIENCES 3208 CAPRICE CT. FORT WAYNE, IN 46808 TEL: (260) 485-4655
1.5 EMERGENCY PHONE NUMBER	(800) 424-9300 CHEMTREC®

SECTION 2 HAZARD IDENTIFICATION

GENERAL HAZARD STATEMENT

To the best of our knowledge, this Safety Data Sheet is prepared to comply with US OSHA 29 CFR 1910.1200 and other governmental agencies, both domestic and foreign. Hazard and precautionary statements are based on maximum listed concentrations of each hazardous ingredient. Unlisted ingredients are not “hazardous” per the OSHA Hazard Communication Standard (29 CFR 1910.1200) and/or other governmental agencies

2.1 CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

GHS CLASSIFICATION IN ACCORDANCE WITH (US) OSHA (29 CFR 1910.1200)

- Acute toxicity, inhalation, Category 4
- Acute toxicity, oral, Category 4
- Eye damage / irritation, Category 2B
- Skin corrosion / irritation, Category 2C

2.2 GHS LABEL ELEMENTS, INCLUDING PRECAUTIONARY STATEMENTS

PICTOGRAM



IRRITANT

SIGNAL WORD **WARNING**

HAZARD STATEMENTS

H302	Harmful if swallowed.
H316	Causes mild skin irritation.
H320	Causes eye irritation.

SECTION 2 HAZARD IDENTIFICATION – continued

PRECAUTIONARY STATEMENTS

- P262 Do not get in eyes, on skin, or on clothing.
 P264 Wash hands thoroughly after handling.
 P270 Do not eat, drink or smoke when using this product.
 P280 Wear protective gloves/protective clothing/eye protection/face protection.
 P301+312 IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell.
 P301+330+331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
 P303+361+353 IF ON SKIN: (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
 P305+351+338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.
 P314 Get medical advice/attention if you feel unwell.
 P332+313 If skin irritation occurs: Get medical advice/attention.
 P337+313 If eye irritation persists: Get medical advice/attention.
 P362+364 Take off contaminated clothing and wash it before reuse.
 P501 Dispose of contents/container according to federal, state, and local regulations.

SECTION 3 COMPOSITION / INFORMATION ON INGREDIENTS

3.1 MIXTURES

Components

1. Sulfuric acid

Concentration 0 - 5 % (weight)

Other names / synonyms SULFURICACID; Sulphuric acid; sulphuric acid

EC no. 231-639-5

CAS no. 7664-93-9

Index no. 016-020-00-8

- Skin corrosion/irritation, Cat. 1A

H314 Causes severe skin burns and eye damage

SECTION 4 FIRST AID MEASURES

4.1 DESCRIPTION OF FIRST AID MEASURES

General Advice	Consult a physician. Show this safety data sheet to the doctor in attendance.
If Inhaled	Remove person to fresh air and keep comfortable for breathing. Call a poison center or doctor if you feel unwell
In Case of Skin Contact	Take off immediately all contaminated clothing. Rinse skin with water/shower for at least 15 minutes. Call a poison center or doctor if irritation develops or persists. Wash contaminated clothing before reuse. Acute and delayed symptoms and effects: Causes mild skin irritation. Signs/symptoms may include localized redness, itching.

**SECTION 4 FIRST AID MEASURES – Continued**

In Case of Eye Contact	Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention/advice. Acute and delayed symptoms and effects: Causes eye irritation. Signs/symptoms may include redness.
If Swallowed	Rinse mouth. Do NOT induce vomiting, call a poison center or doctor. If vomiting occurs naturally, have victim lean forward to reduce the risk of aspiration. Never give anything by mouth to an unconscious person. Acute and delayed symptoms and effects: May cause gastrointestinal irritation. Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

4.2 MOST IMPORTANT SYMPTOMS/EFFECTS, ACUTE AND DELAYED

The most important known symptoms and effects are described in the labelling (See Section 2) and/or in Section 11.

4.3 INDICATION OF IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT NEEDED, IF NECESSARY

No data available.

SECTION 5 FIRE FIGHTING MEASURES**5.1 SUITABLE EXTINGUISHING MEDIA**

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 SPECIFIC HAZARDS ARISING FROM THE CHEMICAL

Sulfuric acid: No data available.

5.3 SPECIAL PROTECTIVE ACTIONS FOR FIRE FIGHTERS

Avoid any skin contact. Effects of contact or inhalation may be delayed. Fire may produce irritating, corrosive and/or toxic gases. Runoff from fire control or dilution water may be corrosive and/or toxic and cause pollution. Wear positive pressure self-contained breathing apparatus (SCBA). Wear chemical protective clothing that is specifically recommended by the manufacturer. It may provide little or no thermal protection. Structural firefighters' protective clothing provides limited protection in fire situations ONLY; it is not effective in spill situations where direct contact with the substance is possible.

Further information: Use water spray to cool unopened containers.

SECTION 6 ACCIDENTAL RELEASE MEASURES**6.1 Personal precautions, protective equipment and emergency procedures**

Wear respiratory protection if necessary. Avoid breathing gas, mist, vapors, or spray. Ensure adequate ventilation. Evacuate personnel to safe areas. For personal protection see section 8.

SECTION 6 ACCIDENTAL RELEASE MEASURES

6.3 Methods and materials for containment and cleaning up

Stop leak if you can do it without risk. Sweep up and shovel into suitable containers for disposal.

Reference to other sections

For disposal see section 13.

SECTION 7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Do not swallow. Wash thoroughly after handling. Do not eat, drink or smoke when using this product. See Section 8 for information on Personal Protective Equipment.

7.2 Conditions for safe storage, including any incompatibilities

Store away from incompatible materials. See Section 10 for information on Incompatible Materials. Keep out of the reach of children.

Specific end use(s)

Apart from the uses mentioned in section 1 no other specific uses are stipulated.

SECTION 8 EXPOSURE CONTROLS / PERSONAL PROTECTION

8.1 Control parameters

1. Sulfuric acid (CAS: 7664-93-9 EC: 231-639-5)

PEL (Inhalation): 1 mg/m³ (OSHA)

PEL (Inhalation): 0.1 mg/m³, (ST) 3 mg/m³ (Cal/OSHA)

REL (Inhalation): 1 mg/m³; USA (NIOSH)

TLV® (Inhalation): 0.2 mg/m³, (Thor.); USA (ACGIH)

TWA (Inhalation): 0.2 mg/m³; USA (ACGIH)

USA. ACGIH Threshold Limit Values (TLV)

TWA (Inhalation): 1 mg/m³; USA (OSHA)

8.2 Appropriate engineering controls

Use ventilation adequate to keep exposures (airborne levels of dust, fume, vapor, gas, etc.) below recommended exposure limits.

8.3 Individual protection measures, such as personal protective equipment (PPE)

PICTOGRAMS



Eye/face protection

If splashes are likely to occur Wear tightly fitting safety goggles. Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

SECTION 8 EXPOSURE CONTROLS / PERSONAL PROTECTION – continued

Skin protection

Wear protective gloves. Consult manufacturer specifications for further information.

Body protection

Wear protective clothing. Clothing with full length sleeves and pants should be worn. The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multipurpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

Thermal hazards

No data available.

Environmental exposure controls

Do not let product enter drains.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

INFORMATION ON BASIC PHYSICAL AND CHEMICAL PROPERTIES

Appearance, form (physical state, color, etc.)	Clear color liquid
Odor	Slight odor
Odor Threshold	No data available
pH	< 1.00
Melting point / freezing point	Approx. 32°F
Initial boiling point and boiling range	212°F +/-
Flash point	No data available
Evaporation rate	1.00
Flammability (solid, gas)	Non-flammable
Upper/lower flammability limits	No data available
Upper/lower explosive limits	No data available
Vapor pressure	Same as water
Vapor density	No data available
Relative density	No data available
Solubility	Completely soluble in water
Partition coefficient: n-octanol/water	No data available
Auto-ignition temperature	No data available
Decomposition temperature	No data available
Viscosity	No data available
Explosive properties	No data available
Oxidizing properties	No data available

SECTION 10 STABILITY AND REACTIVITY

10.1 STABILITY AND REACTIVITY

Contact with incompatible materials. Sources of ignition. Exposure to heat.

10.2 CHEMICAL STABILITY

Stable under normal storage conditions.

10.3 POSSIBILITY OF HAZARDOUS REACTIONS

No data available.

10.4 CONDITIONS TO AVOID

Heat, flames, and sparks. Incompatible products. Keep away from open flames, hot surfaces, and sources of ignition.

10.5 INCOMPATIBLE MATERIALS

Sulfuric acid: Bases, Halides, Organic materials, Carbides, fulminates, Nitrates, picrates, Cyanides, Chlorates, alkali halides, Zinc salts, permanganates, e.g., potassium permanganate, Hydrogen peroxide, Azides, Perchlorates., Nitromethane, phosphorous, Reacts violently with: cyclopentadiene, cyclopentanone oxime, nitroaryl amines, hexalithium disilicide, phosphorous(III) oxide, Powdered metals

10.6 HAZARDOUS DECOMPOSITION PRODUCTS

No data available on product. In the event of fire: see section 5

Sulfuric acid: Hazardous decomposition products formed under fire conditions. - Sulfur oxides

SECTION 11 TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute toxicity

Likely Routes of Exposure: Eye contact. Skin contact. Inhalation. Ingestion.

Components:

Sulfuric acid: LD50 Oral - Rat - 2140mg/kg

Symptoms (including delayed and immediate effects):

Inhalation: May cause respiratory irritation. Signs/symptoms may include burning pain in the nose and throat, coughing.

Ingestion: May cause gastrointestinal irritation. Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

Skin corrosion/irritation

Causes mild skin irritation. Signs/symptoms may include localized redness, swelling, itching.

Serious eye damage/irritation

Causes eye irritation. Signs/symptoms may include cloudy appearance of the cornea, severe pain, tearing.

SECTION 11 TOXICOLOGICAL INFORMATION – continued**Respiratory or skin sensitization**

No data available for product.

Component:

Sulfuric acid: LC50 Inhalation - Mouse - 320mg/m³

Sulfuric acid: LC50 Inhalation - Rat - 510mg/l

Germ cell mutagenicity: No data available.

Carcinogenicity

This product is or contains a component that is not classifiable as to its carcinogenicity based on its IARC, ACGIH, NTP, or EPA classification

OSHA: No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.

Reproductive toxicity

No data available.

Summary of evaluation of the CMR properties

No data available.

STOT-single exposure

No data available.

STOT-repeated exposure

No data available.

Aspiration hazard

No data available.

Additional information: No data available.

SECTION 12 ECOLOGICAL INFORMATION**Toxicity**

No data available on product

Components

Sulfuric acid: LC50 - Daphnia magna (water flea) - 42mg/l - 96hr

Sulfuric acid: EC50 - Lepomis macrochirus (bluegill) - 49mg/l - 48hr

Persistence and degradability

No data available on product

Bioaccumulative potential

No data available on product

Mobility in soil

No data available.

Results of PBT and vPvB assessment

PBT/vPvB assessment not available as chemical safety assessment not required/not conducted

Other Adverse Effects: No data available.

SECTION 13 DISPOSAL CONSIDERATIONS

.Disposal of the product

Disposal should be in accordance with applicable Federal, State and local laws and regulations. Local regulations may be more stringent than State or Federal requirements.

Disposal of contaminated packaging

Dispose of as unused product.

Waste treatment

Contact your supplier or a licensed contractor for detailed recommendations. Follow applicable federal, state and local regulations.

Sewage disposal

The product should not be allowed to enter drains, water courses or the soil.

Other disposal recommendations

Triple rinse the empty containers with water before disposal to reconditioned or land fill or garbage.

SECTION 14 TRANSPORT INFORMATION

DOT (US)

UN Number: NA

Class: NA

Packing Group: NA

Proper Shipping Name: NON-REGULATED MATERIALS

Reportable quantity (RQ): NA

Marine pollutant: No

Poison inhalation hazard: No

IMDG

UN Number:

Class:

Packing Group:

EMS Number:

Proper Shipping Name:

IATA

UN Number:

Class:

Packing Group:

Proper Shipping Name:

SECTION 15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations specifically for the product in question

SARA 302 Components

The following components are subject to reporting levels established by SARA Title III, Section 302:

Sulfuric acid: CAS number: 7664-93-9

SECTION 15 REGULATORY INFORMATION – continued

SARA 311/312 Hazards

Chronic Health Hazard

SARA 313 Components

The following components are subject to reporting levels established by SARA Title III, Section 313:

Sulfuric acid: CAS number: 7664-93-9

Toxic Substances Control Act (TSCA) Inventory

All components of this product are in compliance with the inventory listing requirements of the U.S. Toxic Substances Control Act (TSCA) Chemical Substance Inventory

Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA):

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) RQ: 1,000 lb.,

Contains: Sulfuric Acid

California Prop. 65 Components

WARNING! This product contains a chemical known to the State of California to cause cancer.

Sulfuric acid: CAS number: 7664-93-9

Massachusetts Right To Know Components

Sulfuric acid: CAS number: 7664-93-9

New Jersey Right To Know Components

Sulfuric acid: CAS number: 7664-93-9

Pennsylvania Right To Know Components

Sulfuric acid: CAS number: 7664-93-9

HMIS RATING

HEALTH	1
FLAMMABILITY	0
PHYSICAL HAZARD	0
PERSONAL PROTECTION	B

NFPA RATING

HEALTH	1
FLAMMABILITY	0
INSTABILITY HAZARD	0
SPECIAL HAZARD	W

SECTION 16 OTHER INFORMATION

16.1 Further information/disclaimer

DISCLAIMER: The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigation to determine the suitability of information for their particular purposes. In no event shall manufacturer be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages, whatsoever arising, even if manufacturer has been advised of the possibility of such damages. Date of Preparation/Revision: 7/2024